

AZ-104 – Lab 09a: Implement Web Apps

(Final Revision Notes + Interview Q&A + GitHub Documentation)

◆ Lab Objective (Simple Language)

This lab teaches how to:

- Host a **web application** using **Azure App Service (PaaS)**
 - Deploy code from **GitHub**
 - Use **deployment slots** for safe releases
 - Perform **slot swapping**
 - Understand **scaling limitations** based on pricing plans
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Core Azure Services Used

Service	Purpose
Azure App Service	Host web apps (PaaS)
App Service Plan	Defines compute, pricing & scaling
Deployment Slots	Zero-downtime deployments
Deployment Center	CI/CD integration
GitHub	Source code repository

◆ Task 1: Create Azure Web App

What you did

- Created a **Web App** using **PHP runtime**
- Selected **Linux OS**
- Used **Standard S1 plan** (due to Premium quota limits)

Key concepts

- **Web App ≠ VM**
Azure manages OS, patching, runtime.

- **App Service Plan** controls:
 - CPU & Memory
 - Scaling
 - Slot availability

Important exam point

**Deployment slots are NOT available on Free or Basic plans
Standard and above are required**

◆ Task 2: Deployment Slot (Staging)

What you did

- Created a **staging slot**
- Did NOT clone production settings

Why slots matter

- Test code before production
- Avoid downtime
- Rollback easily

Interview line

Deployment slots allow blue-green deployments in Azure App Service.

◆ Task 3: Deployment Center (GitHub Integration)

What you did

- Forked **php-docs-hello-world**
- Configured:
 - Source: External Git
 - Repo: Your fork
 - Branch: master

Why forked repo?

- Best practice

- Full ownership
- No dependency on public repo changes

Important note

- Repo uses **master**, not main
- Azure must match branch name exactly

◆ Task 4: Slot Swap (Staging → Production)

What happens during swap?

Item	Behavior
Code	Swapped
Traffic	Zero downtime
URL	Production remains same
Settings	Slot-specific stay

Why this is powerful

- Safe deployments
- Instant rollback

◆ Task 5: Scaling (Reality vs Lab)

What the lab says

- Automatic scaling

What actually happened

- **Automatic scaling disabled**
- **Rule-based scaling limited**

WHY?

Because:

- **Automatic scaling requires Premium plan**
- You used **Standard S1**

What you still learned

- Scaling depends on **pricing tier**
 - Platform enforces limits (good exam topic)
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Key Concepts Summary (Very Important)

Azure App Service

- PaaS offering
- Supports multiple runtimes
- No server management

App Service Plan

- Defines:
 - Slots
 - Scaling
 - Pricing

Deployment Slots

- Separate environments
- Swap capability
- Zero downtime

Scaling Types

Type	Availability
Manual	All paid plans
Rule-based	Standard+
Automatic	Premium only

Interview Questions & Answers

Q1: What is Azure App Service?

Answer:

A fully managed PaaS for hosting web apps without managing servers.

Q2: Why use deployment slots?

Answer:

To test changes safely and swap to production with zero downtime.

Q3: Difference between staging and production slot?

Answer:

Same infrastructure, different endpoints and configurations.

Q4: Why fork GitHub repo instead of using directly?

Answer:

Forking gives ownership, stability, and prevents external changes.

Q5: Why autoscaling was disabled?

Answer:

Automatic scaling requires Premium App Service plan.

Q6: What happens during slot swap?

Answer:

Code and runtime swap, URLs remain unchanged, traffic unaffected.

Q7: App Service vs VM?

App Service	VM
PaaS	IaaS
No OS management	Full OS control

App Service

Auto scaling

VM

Manual scaling